

Why Did I Choose This Company?

The company I chose is **American Data** (<https://www.american-data.com/>). It provides electronic charts and a financial system uniquely designed for long-term care. I chose this company because I believe their ecosystem can be significantly enhanced by addressing a major administrative bottleneck: volunteer coordination. By integrating an AI-driven matching system with their existing scheduling framework, it can transform volunteer management from an administrative burden into a highly efficient feature that directly improves matching accuracy.

The Problem:

When I was a CEGEP student, I volunteered at the Donald Berman Maimonides Geriatric Centre for two years and did some work in the Jewish General Hospital in Montreal. During this time, I experienced some inefficiencies of traditional volunteer tracking:

- **The Compatibility Gap:** Many volunteers were assigned a floor and left to find residents to engage with. However, due to the sheer volume of residents, their conditions and preferences were poorly communicated through nurses and social workers. Volunteers often waste weeks trying to find compatible residents to have enjoyable conversations with, causing delays and unnecessary stress.
- **The Opportunity Gap:** Clinical researchers and doctors often only advertise their information to university students or post it internally, leaving many high school students who lack online searching experience with fewer opportunities, which could negatively impact their med school applications.
- **Administrative Burden:** Volunteer coordinators spend an excessive amount of time manually writing attestation letters for students needing

proof of their hours and impact. Furthermore, many volunteers are highly skilled medical professionals from countries like India, Iran and Philippines. Even though they have years of experience, the Canadian government requires them to be recertified here. Therefore, writing personalized attestation letters for everyone is not an easy task!

My AI Proposal:

The AI volunteer matching system functions like an internal networking platform, similar to LinkedIn. It focuses purely on pairing volunteers with residents and researchers. To maintain security and privacy, only approved administrators (like volunteer coordinators and primary researchers) can register new volunteers. This is also necessary since every volunteer needs to go through a mandatory background check.

There are three types of profiles:

1. **Volunteers:** Academic goals, hard/soft skills, interest keywords.
2. **Researchers:** Ongoing projects, expected skills, research interests.
3. **Residents (Names Redacted):** Life history, hobbies, languages, cognitive profile, interests.

Why:

Residents' names **need** to be redacted because we have to respect their privacy. Once a volunteer is matched to a resident, they can ask the nurses for their preferred name and prefix.

It is actually very simple to obtain residents' profiles because nurses are often required to assess them on a regular basis. All admins need to do is to remove the names and keep them up to date. **Maybe there can be another automated AI workflow that automatically converts nurses' reports into keywords!**

Core Features:

- **Automated Pairing:**

Layer 1: Utilizes a lightweight Natural Language Processing (NLP) model to map the volunteer's "Interest Keywords" and "Career Goals" against Researcher "Projects" or Resident "Life Histories." This ensures semantic hits even without exact phrasing (matching "neurodegeneration" with "dementia").

Layer 2: Weighted Scoring Matrix

Research/Doctor Matching: Overlap in technical skills (40%), Research interest (40%), Goal alignment (20%).

Care Coordinator/Resident Matching: Keyword/Hobby overlap (50%), Relevant soft skills (30%), Language compatibility (20%).

My reasoning for key design decisions: After some consideration, I removed the constraint filtering for required skills and schedule conflicts because, in my experience, some doctors make exceptions and some don't. In the end, I think it should be up to the volunteer coordinators to decide.

During my 2023 volunteer work at Maimonides, I spent my first few weeks assisting with the occupational therapy coordinator and organized games on my assigned floor. While fulfilling, I struggled to build deeper connections with those residents. This changed when I substituted an absent volunteer on another floor and met a resident who lived in Hong Kong and had an interest in Chinese food. What started as a standard shift turned into hours of continuous conversation and many life lessons. That single interaction transformed my entire volunteering experience. It made me realize that stepping outside assigned routines to find

common interests is crucial, which is why I prioritize matching people based on shared interests over availability and requirements.

Also, assisting with clinical studies isn't as difficult as many imagine. After finishing my regular volunteer program, I saw a poster for an active cognitive study. Since I had free time after school, I worked out a schedule with the research assistant and took on the role. The work itself was straightforward: we would visit residents' rooms and ask simple memory questions, like what year it was or what they had seen on the news earlier. However, opportunities like this are usually hidden from high school students with little experience in online research. Furthermore, even though researchers have tight schedules, people always make time for what's important to them. That's why I made the "Hard Requirement" filters in the matching system to be flexible.

- **AI Attestation Generator (Future Implementation):** Automatically drafts accurate, personalized attestation letters for university applications and certifications based on a volunteer's logged hours, matched roles, and utilized skills.

Technical Approach:

- **Backend:** Python for simplicity and compatibility
- **Frontend UI:** A responsive JavaScript dashboard for coordinators to view the match feed, approve pairings and manage users.
- **NLP Engine:** a lightweight NLP (WordLlama <https://github.com/dleemiller/WordLlama>) maps volunteers' keywords against researchers' interests or residents' experiences, etc. Furthermore, the model understands relationships and connects synonyms, i.e., treating

"dementia" and "memory loss" as linked concepts. This could generate more accurate matches without using exact keyword tags.

Note: I used WordLlama because it runs fast on my laptop. However, there are definitely better NLP/LLM models to use. With better hardware, there could be changes. For example, I may use a vector database to store the semantic terms. It would be a great addition in the future.

- **Scoring Algorithm:** Applies a weighting-based method that compares technical skill overlap, goals, and shared hobbies to rank matches.
- **Attestation Generation (Future Implementation):** Integrates a LLM API instructed via prompt templates to generate letters using data from the database, which improves accuracy and efficiency.

A final note: in the end, maybe we can also make an AI system that matches nurses or private care givers to compatible residents, which could make long-term care units much more alive! This could be very impactful!

Intro Video:

<https://drive.google.com/file/d/11Rb8tiXagPccHbAHpkWJmWxtYTLZakF6/view?usp=sharing>

Github Link of the Prototype:

https://github.com/davezhang1116/volunteer_system